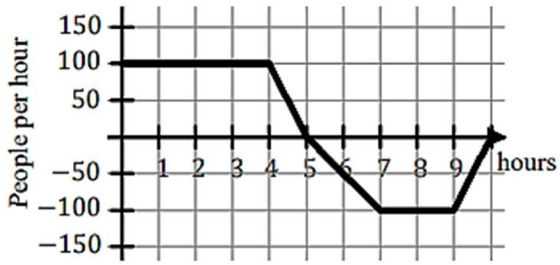
Worksheet 6–1 A Accumulation of Rates #1
AP Calculus

Name_____

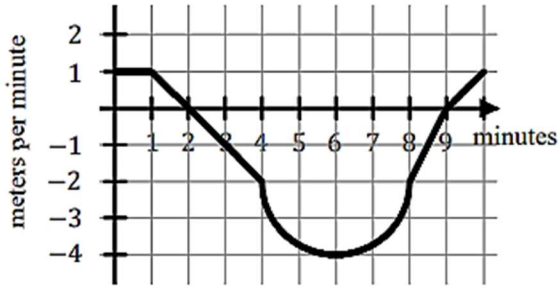
1.) The graph at the right shows the rate of change for the number of people in a museum t hours after it opens.



a.) How many people are in the museum after 5 hours?

b.) How many people are in the museum after 10 hours?

2.) The graph below shows the velocity of a particle moving along the x–axis, measured in meters per minute. At $t = 0$ minutes, the particle is at the origin.

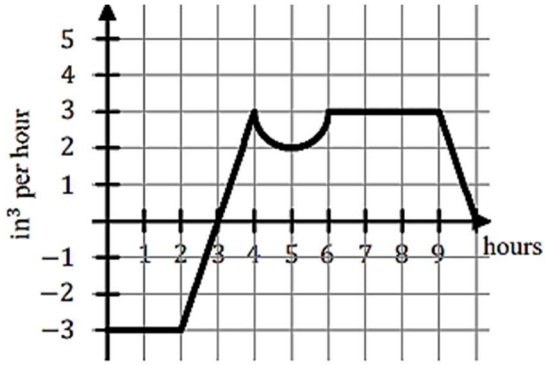


a.) Where is the particle after two minutes?

b.) Where is the particle after six minutes?

c.) Where is the particle after ten minutes?

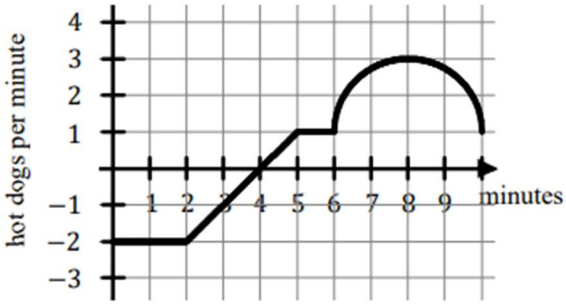
3.) The amount of water in a small pond is changing at the rate modeled in the graph below, where the rate is measured in cubic inches per hour.



a.) How much water has been gained or lost during the first three hours?

b.) How much water has been gained or lost after 10 hours?

4.) The graph below shows the rate at which hot dogs are on Mr. Kelly's plate. Assume there are 10 hot dogs on the plate at $t = 0$ minutes.

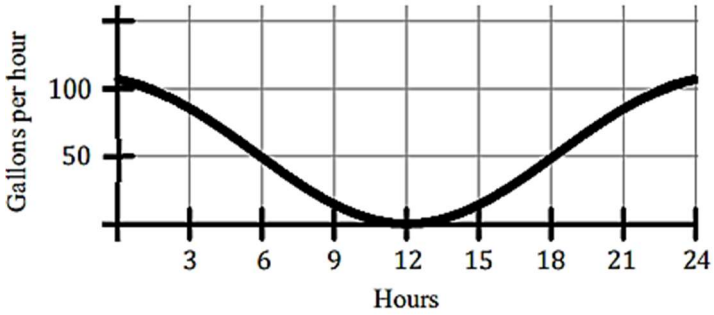


a.) How many hot dogs are on Mr. Kelly's plate after six minutes?

b.) How many hot dogs are on Mr. Kelly's plate after 10 minutes?

5.) The flow of water, in gallons per hour, through a garden house on a hot summer day is given by the graph shown below. Of the following, which best approximates the total number of gallons of water that passed through

(A) 400
(B) 800
(C) 1,200
(D) 2,400



Each function listed represents a rate of change. What are the units for the area under the curve?

6.) $f(t)$ is measured in milligrams per year and t is measured in years.

7.) $g(t)$ is measured in gallons per month and t is measured in months.

8.) $h(t)$ is measured in feet per hour and t is measured in hours.